Biodegradable polymers

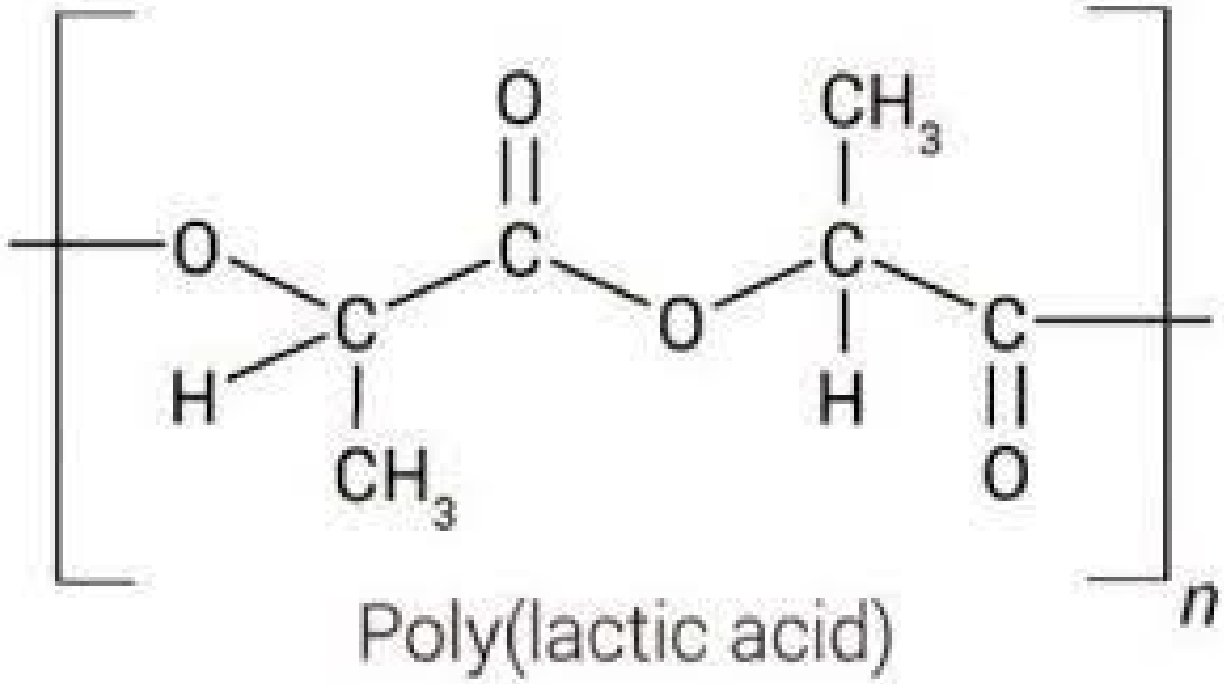
Sustainable Alternatives to traditional polymers

- To address environmental concerns associated with plastic pollution and resource depletion
- **Biodegradable Polymers**: break down into natural components when exposed to environmental conditions, reducing their persistence in the environment.
- **Bio-based Polymers:** derived from renewable resources such as plants, algae, or waste materials. These polymers reduce dependency on fossil fuels.
- **Bioplastics:** a subset of biodegradable and bio-based polymers, often derived from renewable resources and can be composted, recycled, or biodegraded under appropriate conditions.
- **Natural Polymers:** biocompatible, biodegradable, and environmentally friendly, derived from renewable resources such as cellulose, starch, or proteins.
- **Recycled Polymers:** Recycling is an essential part of the circular economy, where materials are reused and recycled to minimize waste and resource consumption. Recycled polymers, also known as post-consumer, recycled (PCR) or post-industrial recycled (PIR) polymers, are produced from reclaimed plastic waste and can be used to manufacture new products.

Biodegradable Polymers

- Polymers that can be broken down by microorganisms, such as bacteria, fungi, and algae, into harmless byproducts like water, carbon dioxide, and methane, under aerobic or anaerobic conditions
- environmentally friendly alternative to traditional plastics
- *Biobased Biodegradable Polymers*: derived from renewable resources such as plants, animals, or microorganisms.
- Eg. Polylactic acid (PLA), polyhydroxyalkanoates (PHA).

- *Synthetic Biodegradable Polymers*: created through chemical synthesis but designed to degrade under specific environmental conditions.
- Eg. polycaprolactone (PCL) and polybutylene succinate (PBS)



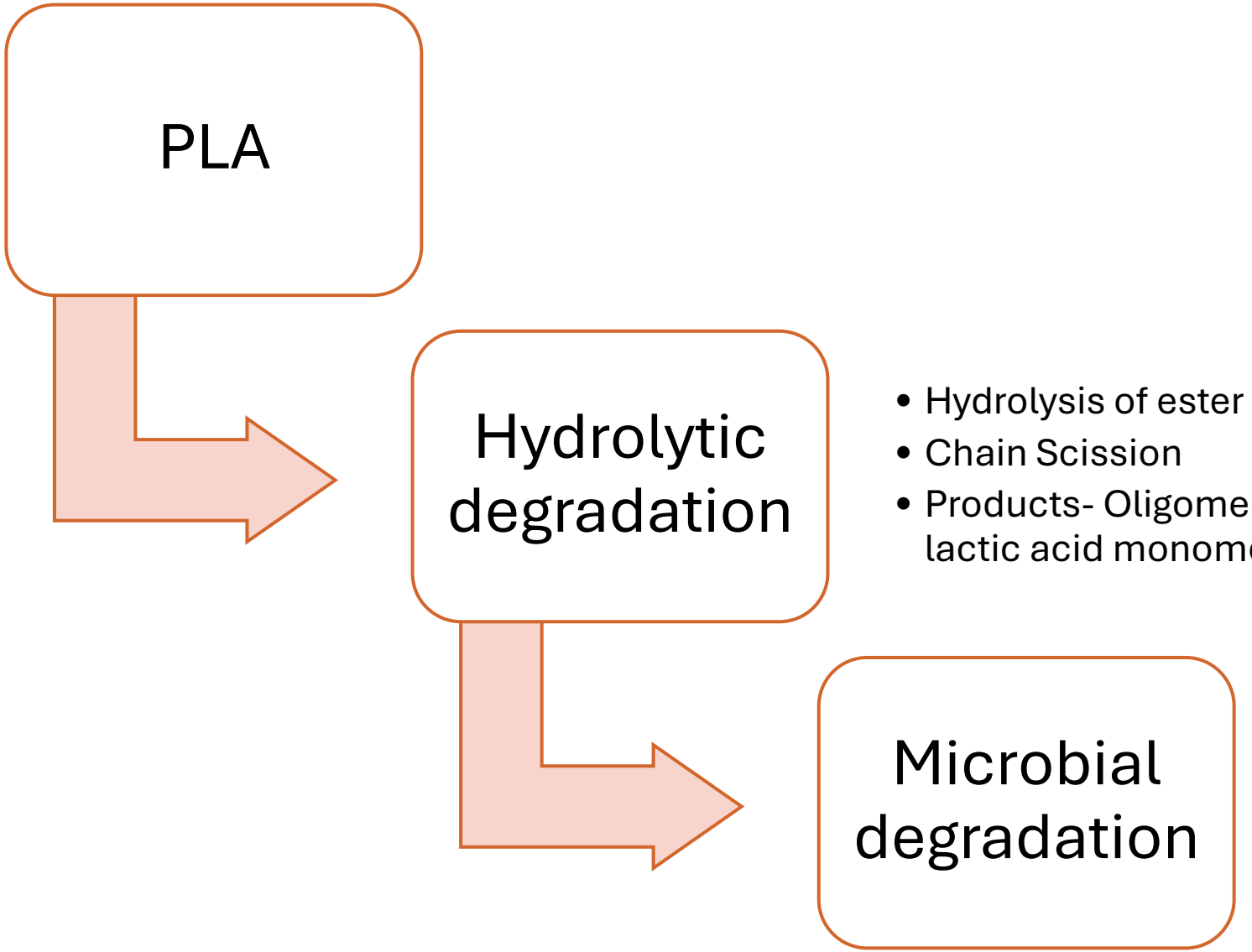
Poly(lactic acid)

Poly(lactic acid) (PLA): biodegradable and bioactive thermoplastic derived from renewable resources such as corn starch, sugarcane, or cassava roots, reducing dependence on fossil fuels.

Mechanism of Degradation of PLA

The degradation of polylactic acid (PLA) involves both physical and chemical processes, primarily hydrolysis, which leads to the breakdown of its polymer chains into smaller, more biodegradable molecules.

Mechanism of Degradation of PLA



- Hydrolysis of ester bond
- Chain Scission
- Products- Oligomers and lactic acid monomers

- End products CO_2, H_2O and microbial biomass

Environmental Conditions

- PLA's degradation rate depends heavily on environmental conditions. In natural environments like soil or water, degradation is slower because of lower temperatures, reduced microbial activity, and limited moisture. In industrial composting settings, which provide optimal conditions (higher temperatures, moisture, and controlled microbial activity), PLA can degrade more quickly, typically within a few months.

Advantages

- Renewable Resources: Made from renewable biomass, reducing reliance on fossil fuels.
- Biodegradability: decomposes into lactic acid, a naturally occurring substance, under composting conditions, reducing environmental pollution.
- Low Toxicity: Safe for food contact and medical applications.

Disadvantages

- Low thermal Resistance
- Brittle and less impact-resistant compared to some traditional plastics.
- Degradation Conditions: Requires industrial composting facilities to biodegrade effectively, which may not be available everywhere.

Uses

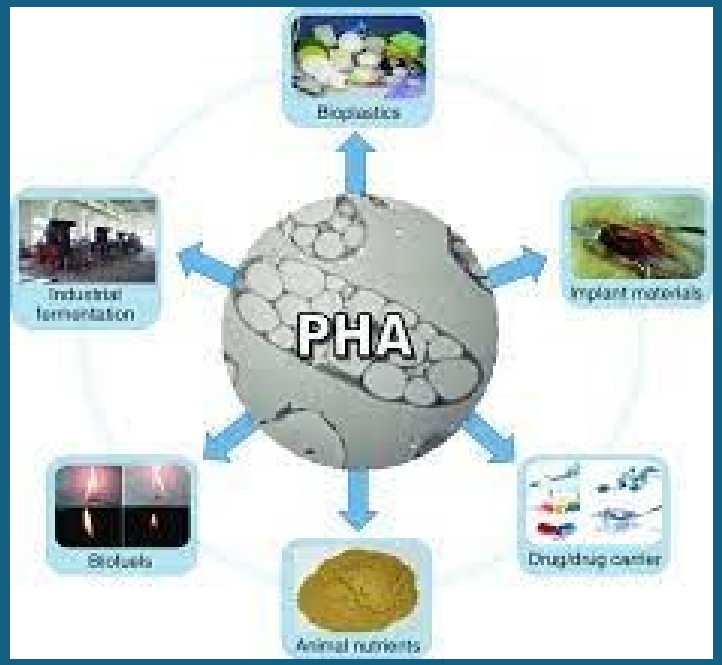
Production of biodegradable plastic films, containers, cups, and utensils.

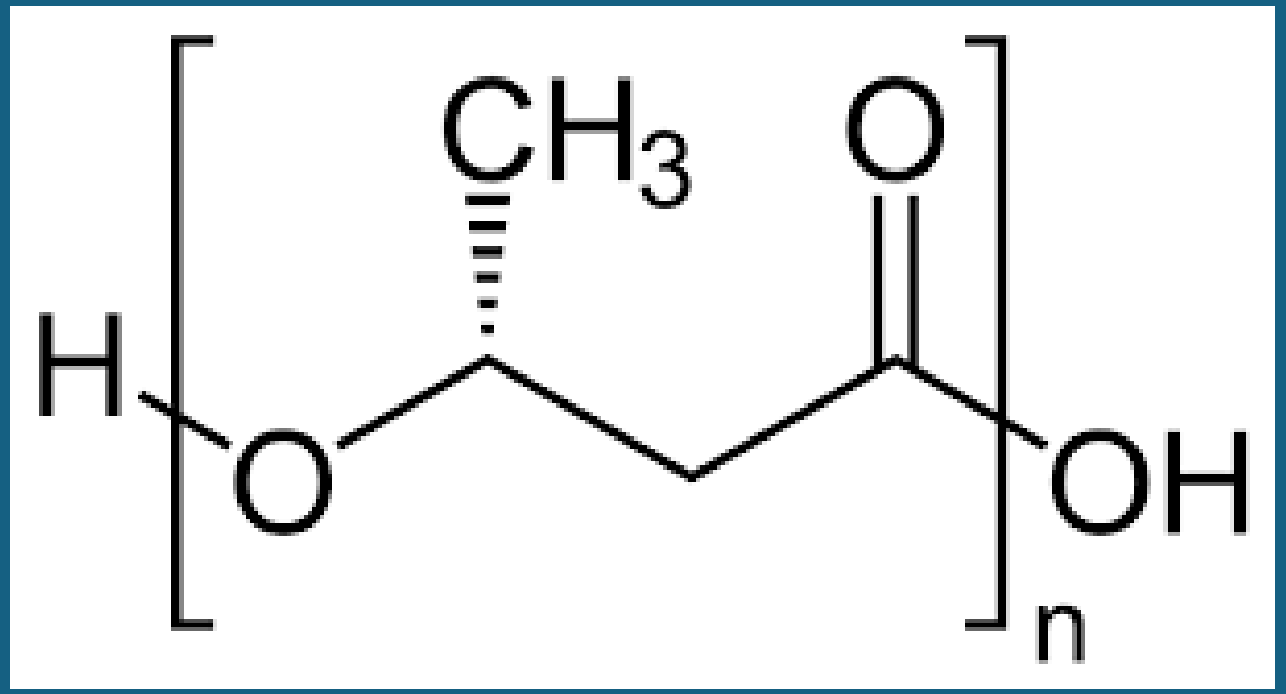
Popular material for 3D printing due to its ease of use, low printing temperature, and minimal warping.

for sutures, stents, and drug delivery systems due to its biocompatibility and biodegradability.

PLA fibers are used in clothing, upholstery, and non-woven fabrics.

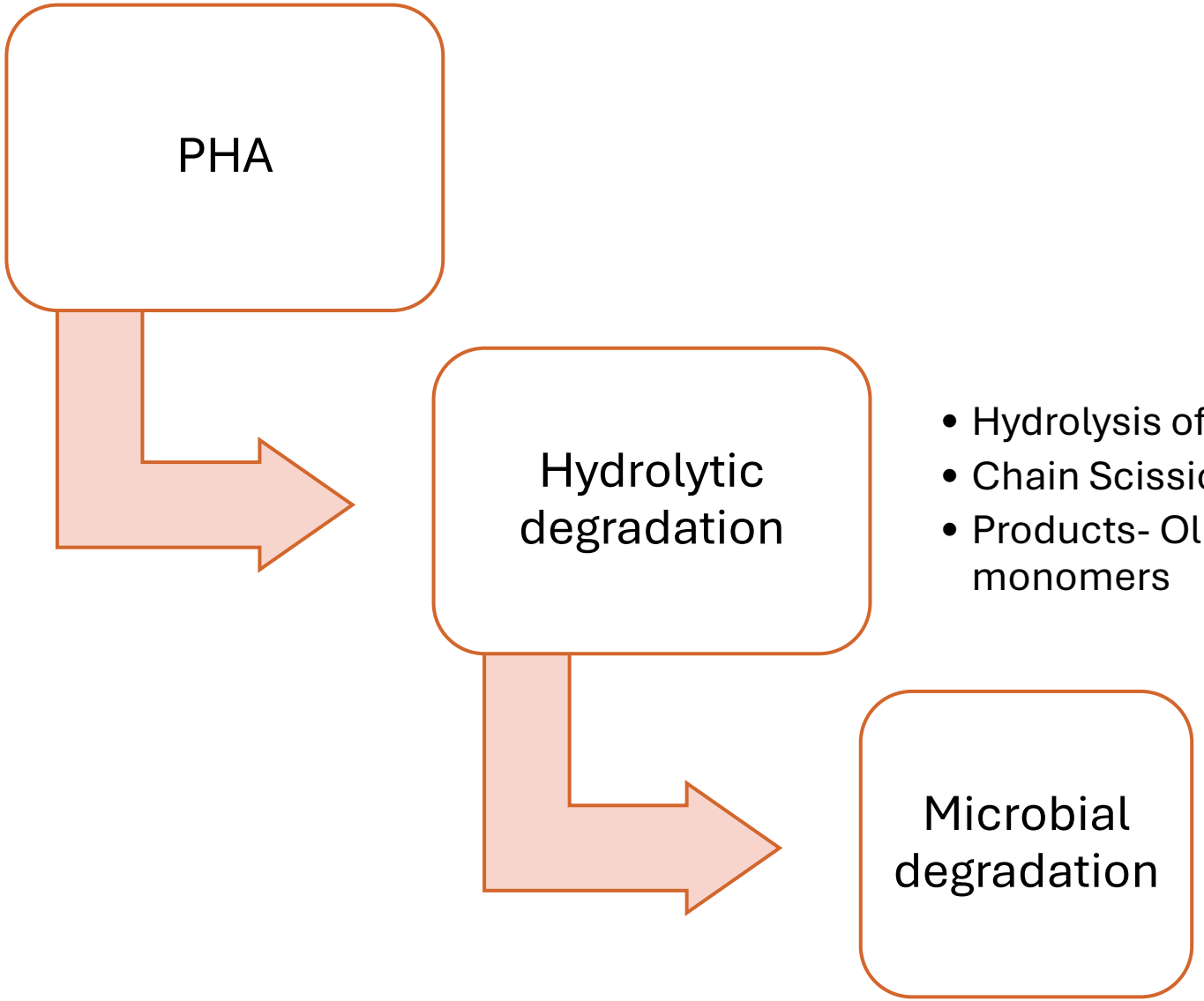
used in mulch films and other agricultural applications.


Bioplastics
Industrial fermentation
PHA
Implant materials
Biofuels
Drug/drug carrier
Animal nutrients


CH3
O
H
O
OH
n

Polyhydroxyalkanoates (PHAs): biodegradable and biocompatible polymers produced by various microorganisms.

Mechanism of Degradation of PHA



- Hydrolysis of ester bond
- Chain Scission
- Products- Oligomers and monomers

- Enzyme – PHA depolymerase break down polymer into smaller monomeric units like hydroxybutyrate (PHB) or hydroxyvalerate (PHBV)
- End products CO_2, H_2O and microbial biomass

Differences in PHA and PLA Degradation:

- **Hydrophobicity:** PHAs are generally more hydrophobic than PLA, so they may not absorb water as readily, making hydrolysis somewhat slower in the initial stages.
- **Microbial Action:** PHAs are more readily utilized by a wide variety of microorganisms compared to PLA, which means the microbial degradation phase is usually faster for PHAs.

Advantages

- Biodegradability: This reduces the accumulation of plastic waste in landfills and oceans.
- Renewable Resources: produced from renewable resources, including agricultural waste, plant oils, and sugars, thus reducing reliance on fossil fuels and contributing to a circular economy.
- Reduced Carbon Footprint: Production of PHAs results in lower greenhouse gas emissions compared to conventional plastics, contributing to climate change mitigation efforts.

Disadvantages

- High Production Cost
- Production Complexity: The production process of PHAs is more complex and energy-intensive, requiring specific microorganisms and conditions.
- Feedstock Limitations: The production of PHAs often relies on specific feedstocks, such as sugar or plant oils, which can lead to competition with food resources.

Uses

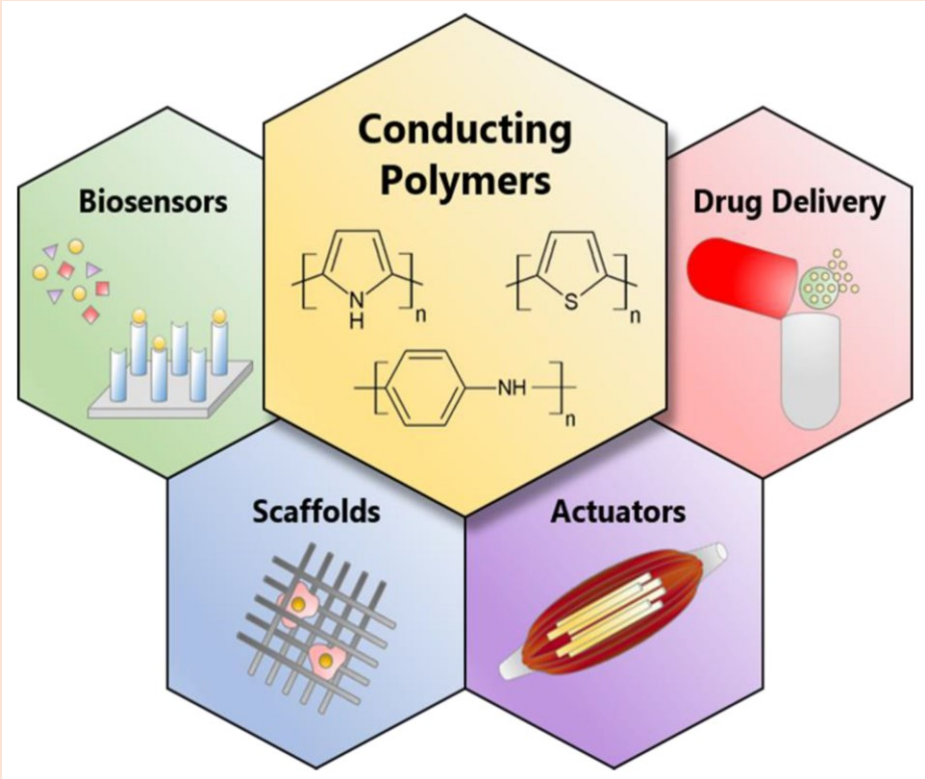
Biocompatibility: non-toxic and biocompatible, making them ideal for medical applications.

Consumer Goods: disposable items, textiles, and personal care products

Agriculture: in agricultural films, controlled-release fertilizer coatings, and biodegradable plant pots, promoting sustainable agricultural practices.

Packaging: production of biodegradable packaging materials, helping to reduce plastic pollution.

CONDUCTING POLYMERS



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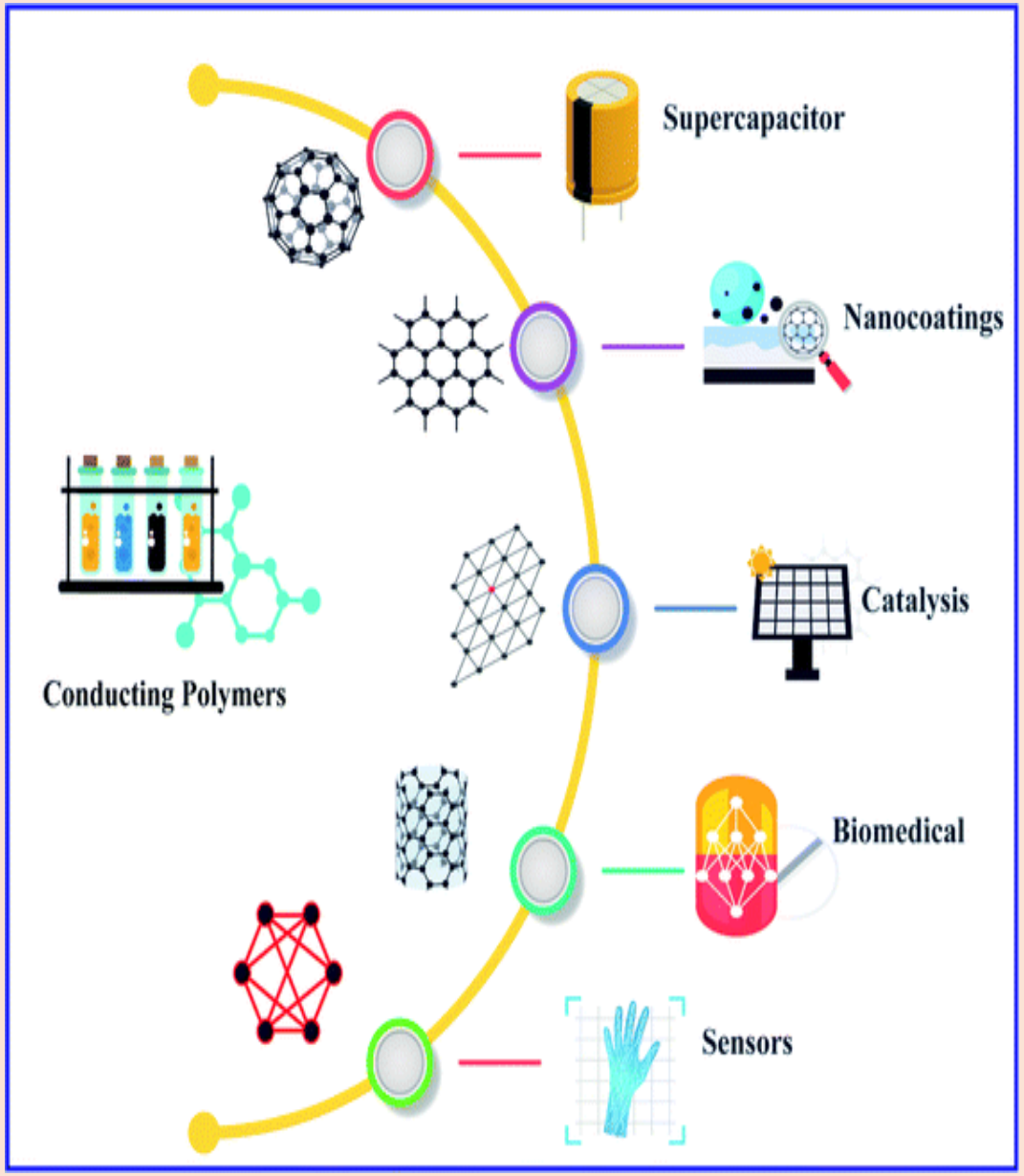

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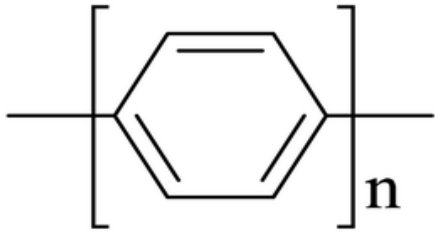
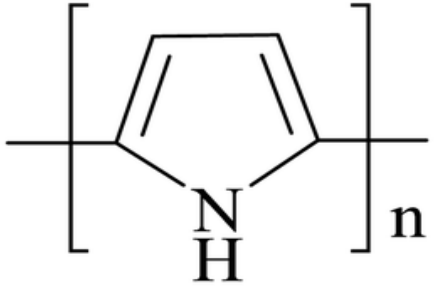
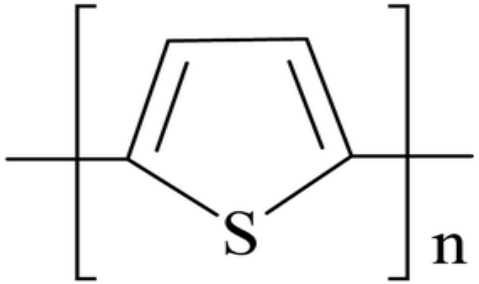
CONDUCTING POLYMERS



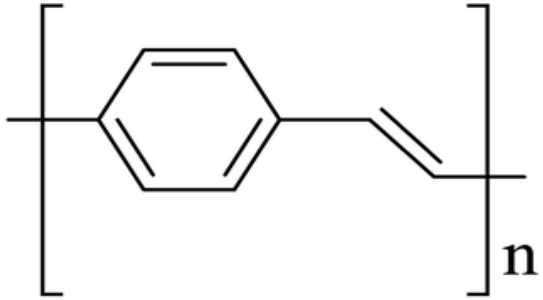
poly(p-phenylene)



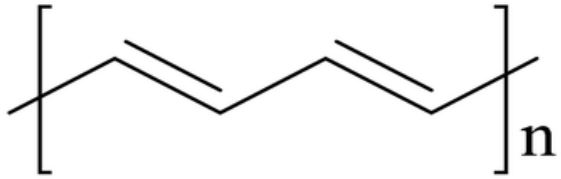
Polypyrrol



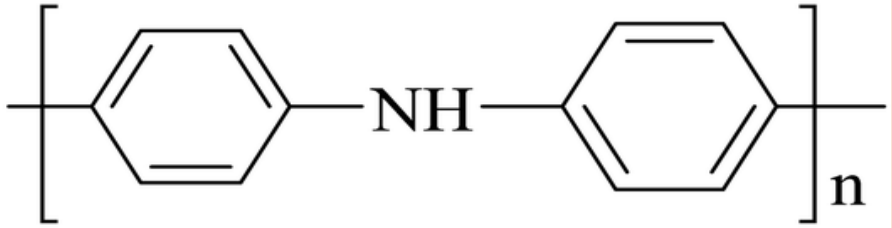
Polythiophene



poly(p-phynylene vinylene)



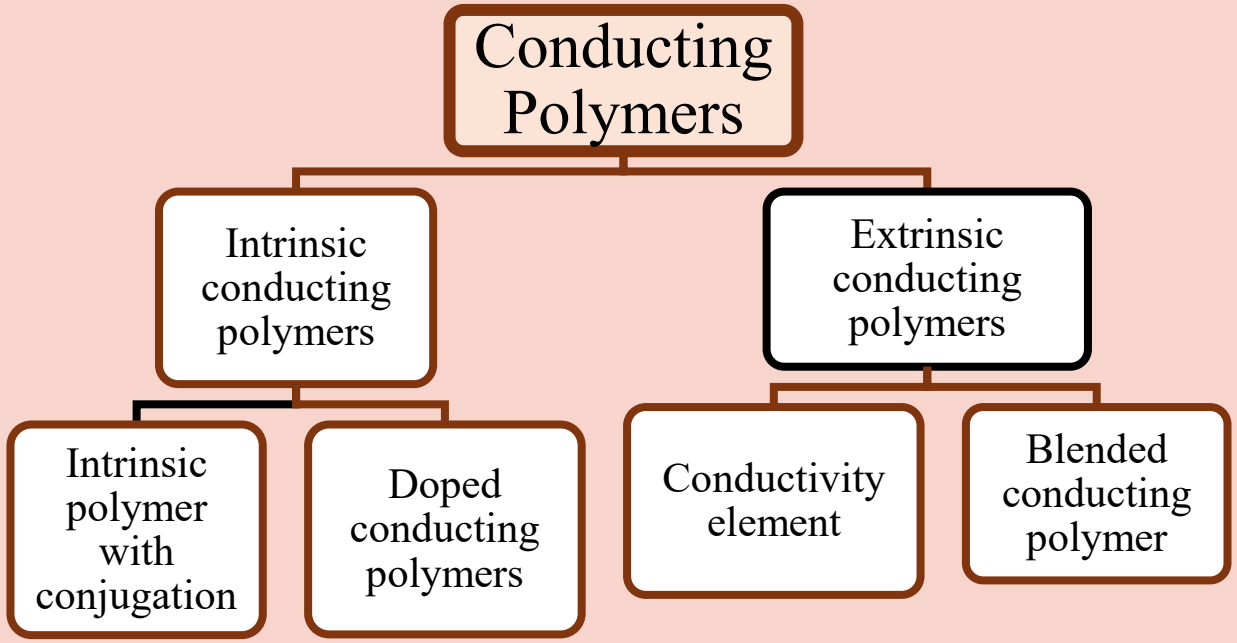
Polyacetylyne



Polyaniline

- Those polymers which conduct electricity are called conducting polymers.
- The conduction of polymers may be due to **unsaturation** or due to the presence of **externally added ingredients** in them.

Classification

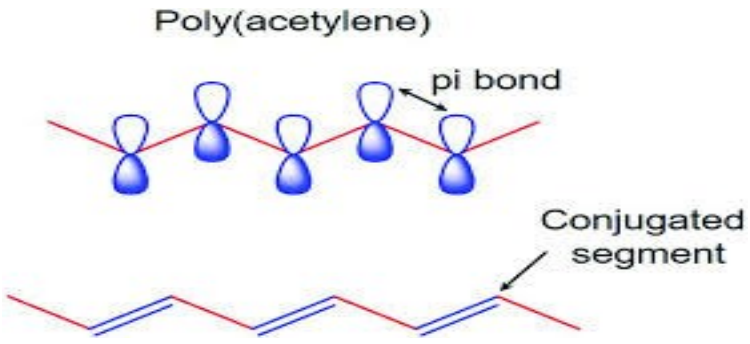


Intrinsic CONDUCTING POLYMERS(ICPs)

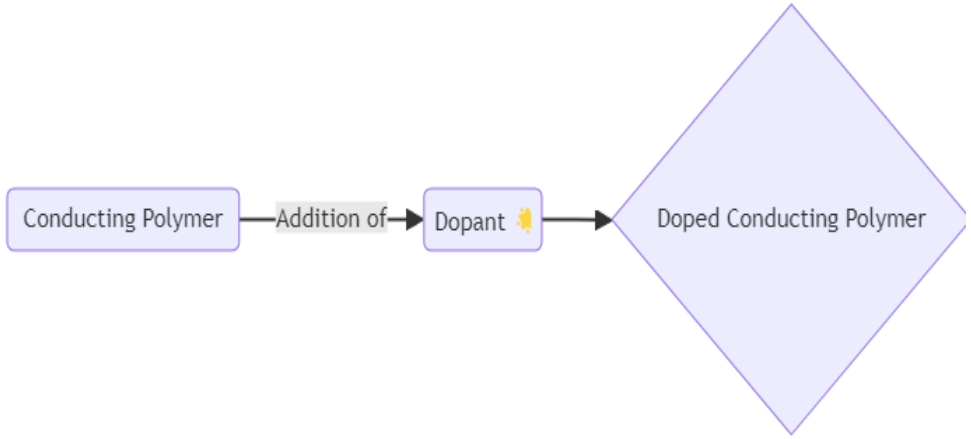
- These polymers are characterized by intensive of double bonds in their structure i.e., backbone of the polymer.

Classification

Conducting polymers having conjugation

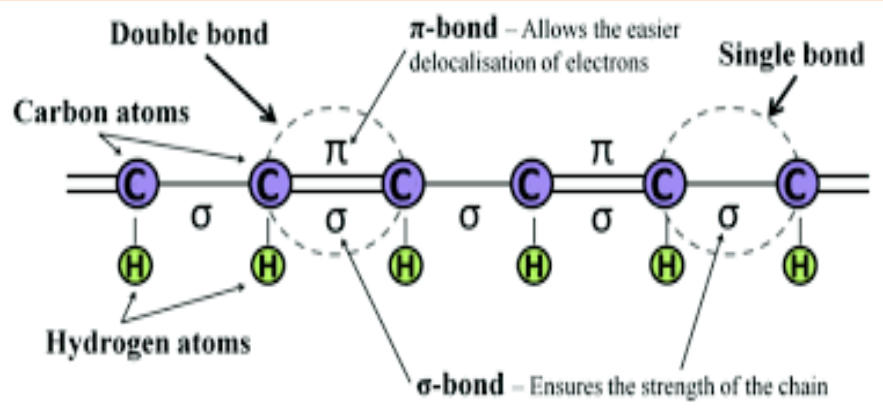


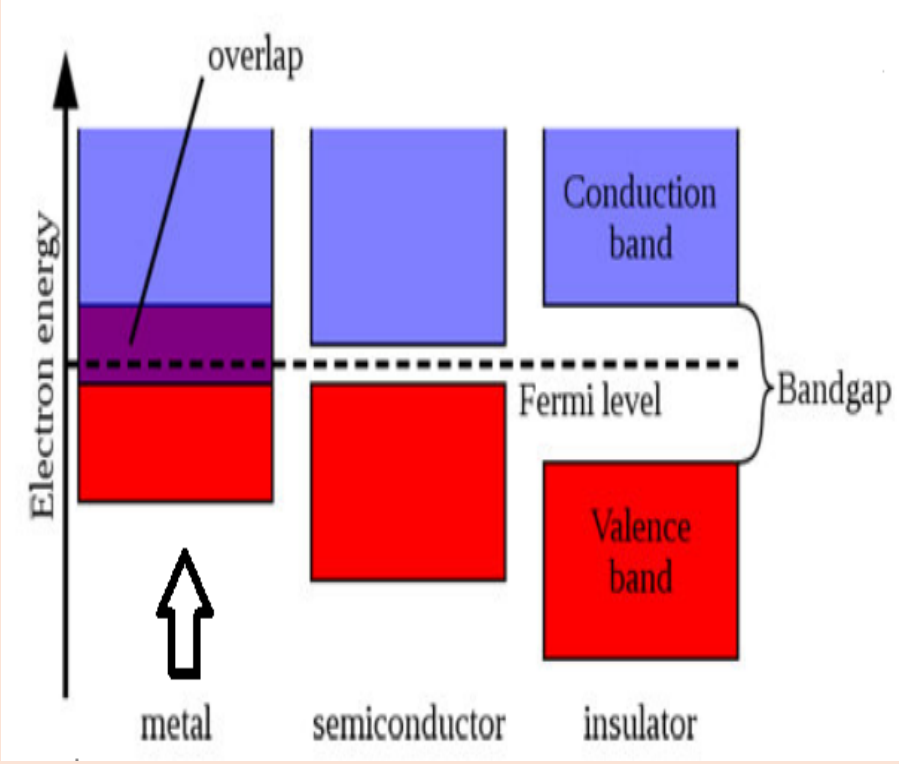
Doped Conducting polymers



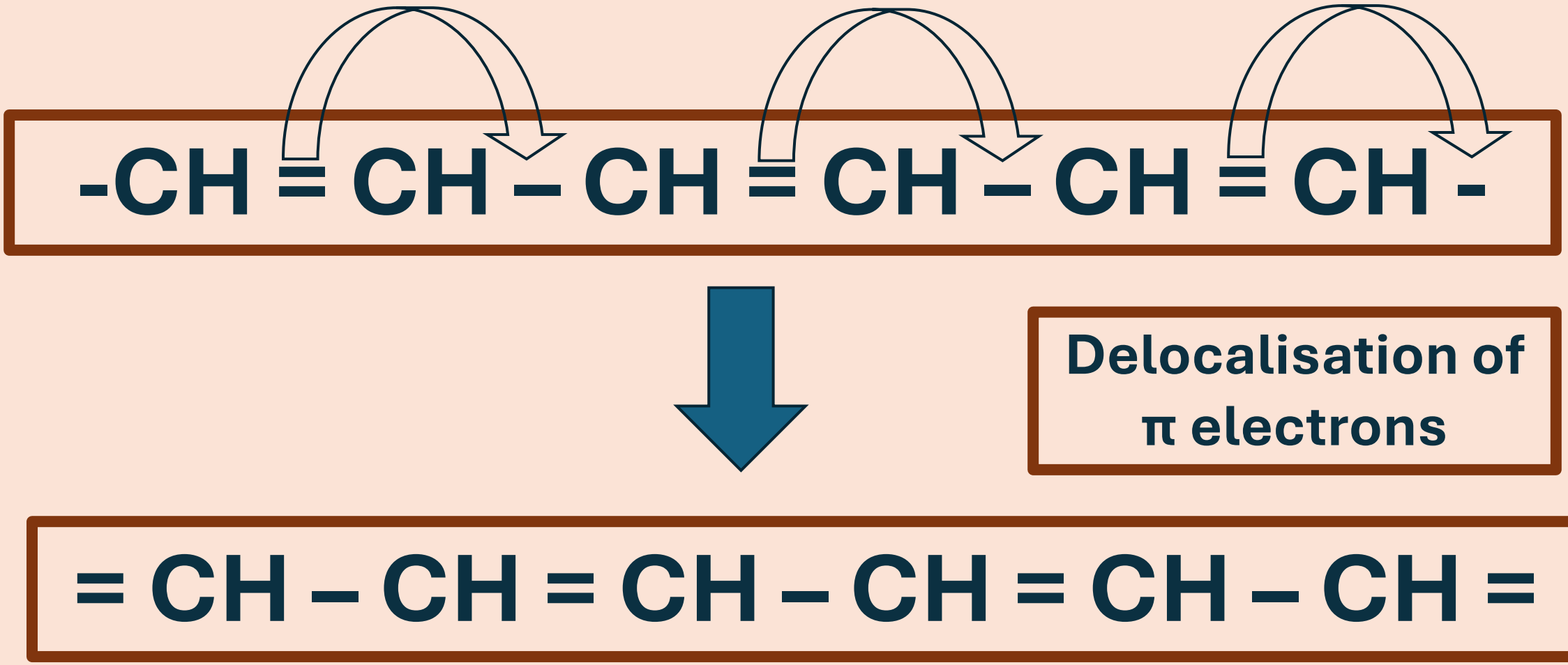
1. Conducting polymers having conjugation

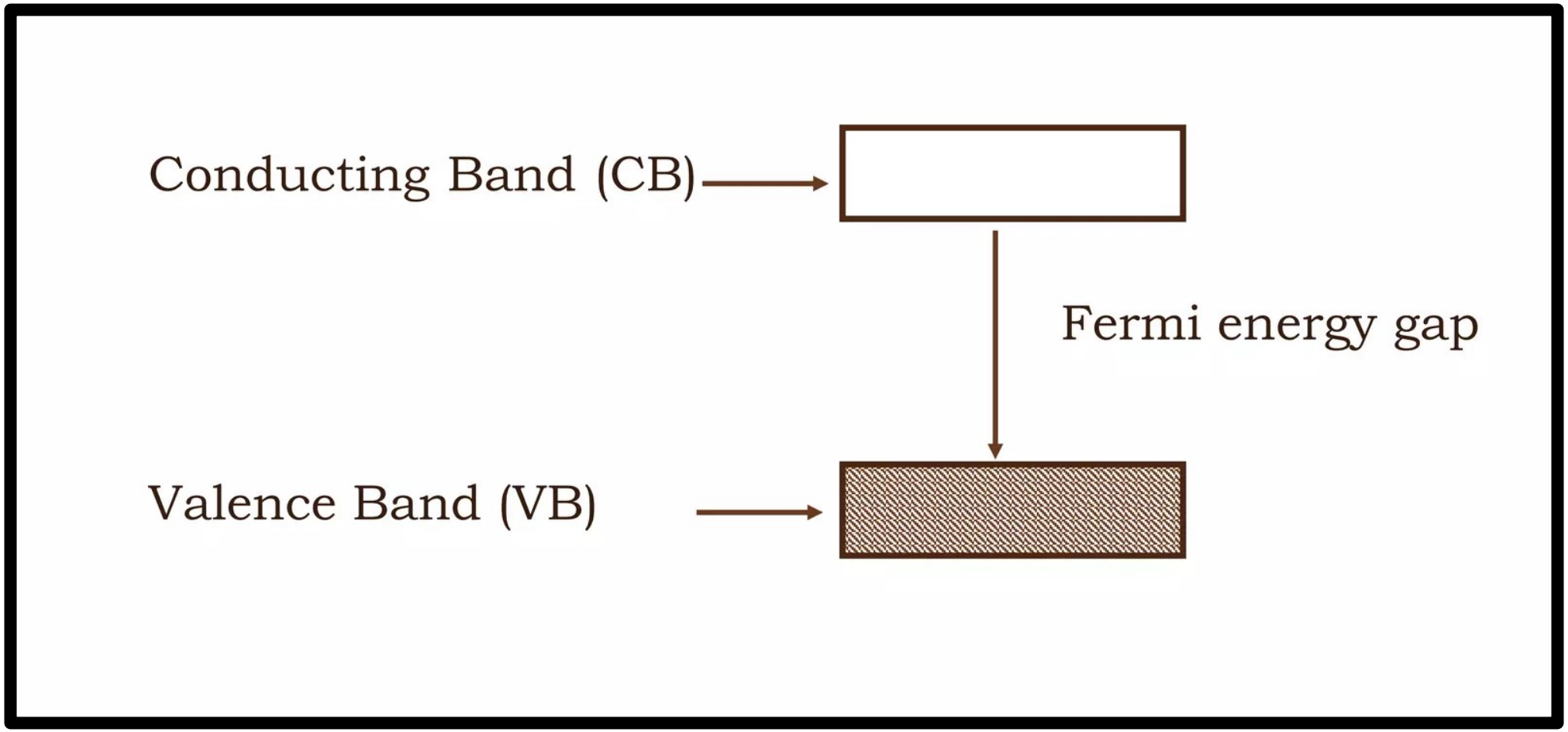
- ❑ Polymers having **conjugated double bonds** in the backbone possess their **conductivity due to π electrons**.
- ❑ In π bonding the overlapping of the orbitals is lateral over the entire backbone resulting in the formation of
  - ✓ **lower energy valence bands and**
  - ✓ **higher energy conducting bands** which were separated by a significant **fermi energy gap**.
- ❑ The electrical conductivity takes place only after thermal or photolytic activation of the electrons, which give them sufficient energy to jump the gap and reach into conduction band.



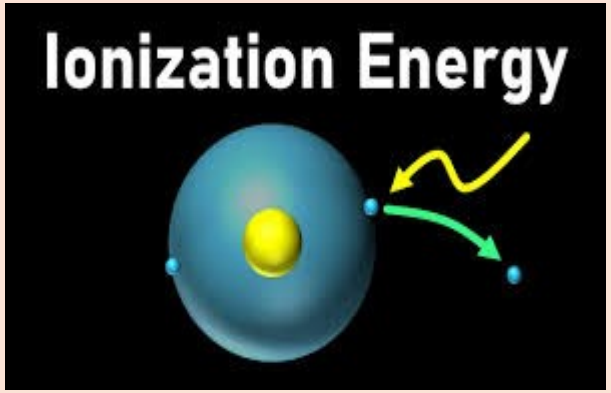


Example: **Polyacetylene**



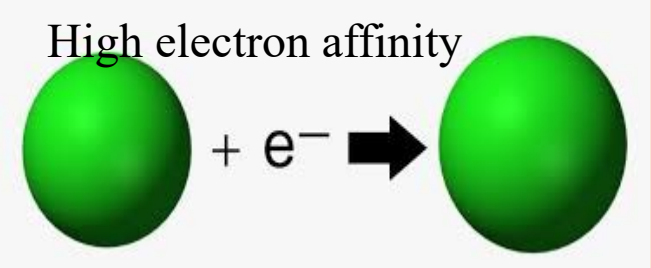

Conducting Band (CB)
Fermi energy gap
Valence Band (VB)

2. Doped conducting polymers



The conducting polymers having π electrons in the backbone can
easily be oxidised or reduced because they possess :

- ✓ **low ionization potential and**
- ✓ **high electron affinities**



Hence their conductance can be
❑increased by introducing a positive charge or negative charge on polymer backbone by oxidation or reduction.

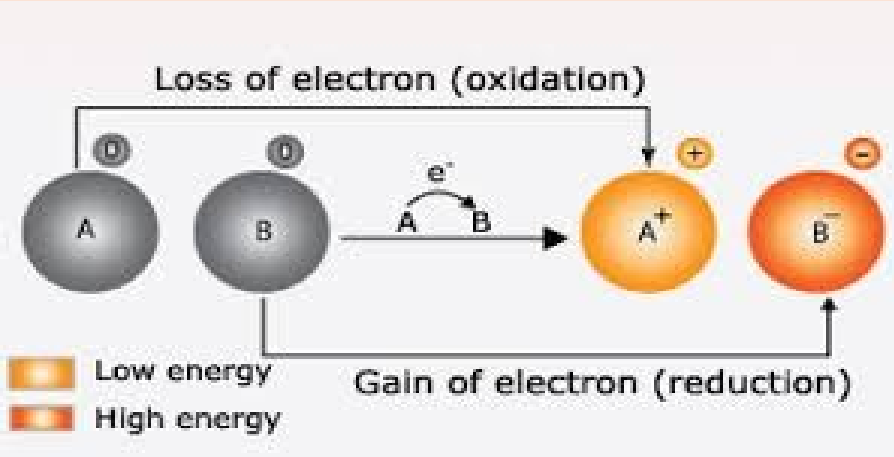
- This process is similar to semiconductor technology and is called **Doping**.

❑Doping is of two types:

- ➢ **Creating a positive site on polymer backbone called p-doping**
- ➢ **Creating a negative site on polymer backbone called n-doping**

p-doping

- p-doping is done by oxidation of a conducting polymer like polyacetylene with Lewis's acid or Iodine vapor



$$-(CH=CH)_n- + 2\,FeCl_3 \longrightarrow -(CH=CH)_n^{+}\ \ FeCl_4^{-} + FeCl_2$$

$$-(CH=CH)_n- + 3\,I_2 \longrightarrow 2-(CH=CH)_n^{+}-I_3^{-}$$

p-doping

- During oxidation process the removal of π electrons from polymer backbone led to the formation of a delocalized radical called ion called **polaron** having a hole in between valence band and conduction band

A polaron is a quasiparticle used in condensed matter physics to describe the interaction between an electron and the lattice of a solid material.

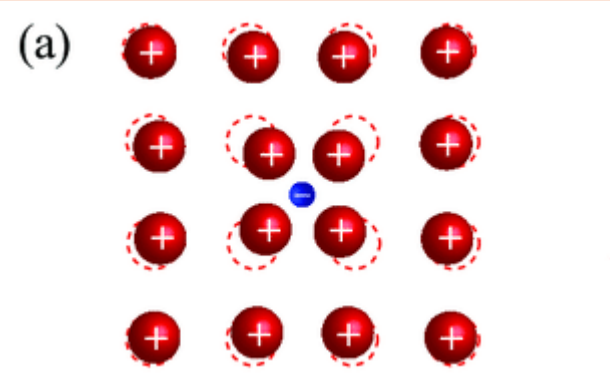
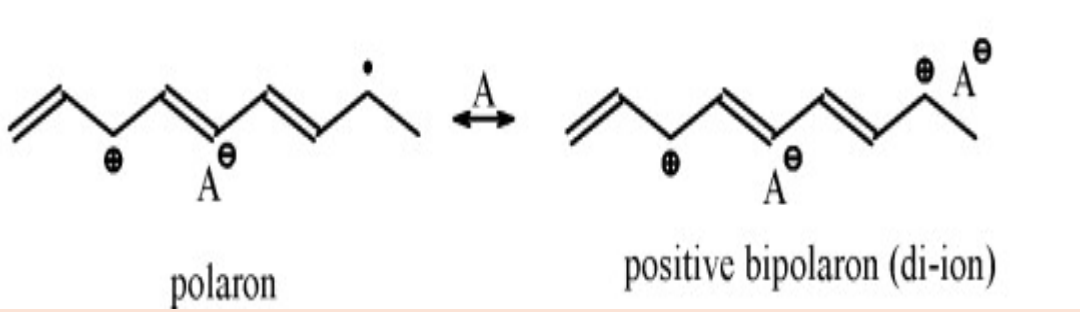


Fig: Polaron: an electron in a solid interacts with the crystal lattice. As a result, it is 'dressed' by the resulting polarization cloud of nuclear displacements and forms the so-called polaron.

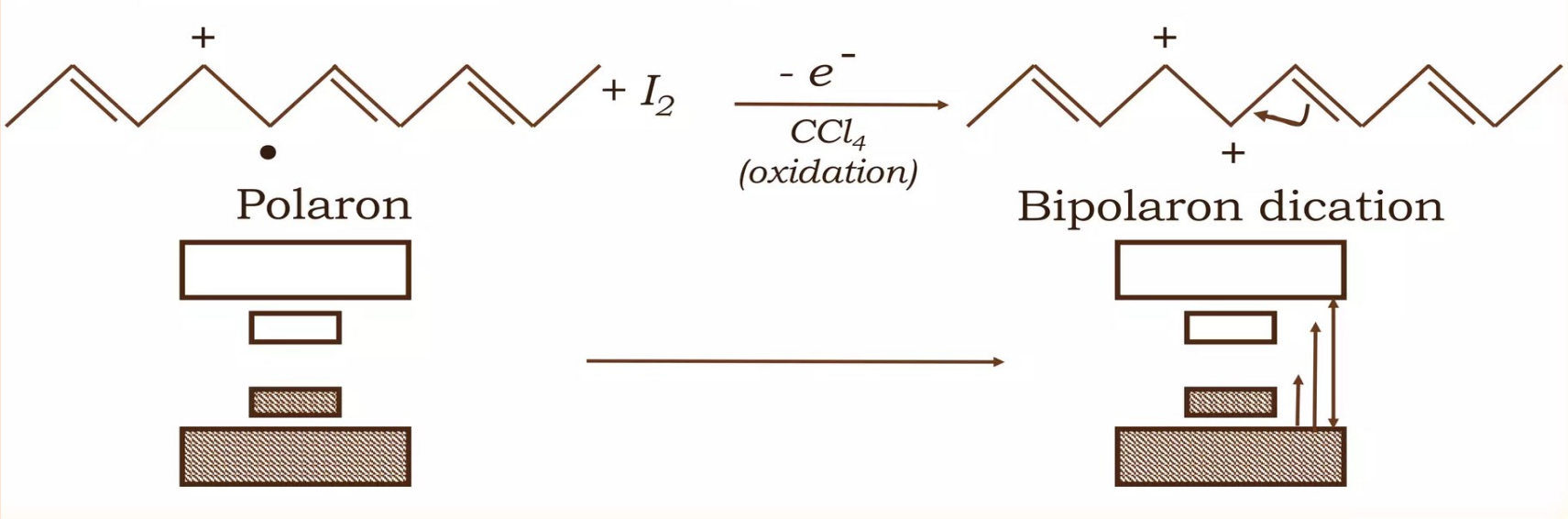
p-doping

- The second oxidation of the polaron results in the two positive charge carriers in each chain called **bipolaron**, which are mobile due to delocalization.
- These delocalized charge carriers are responsible for conductance when placed in electric field.

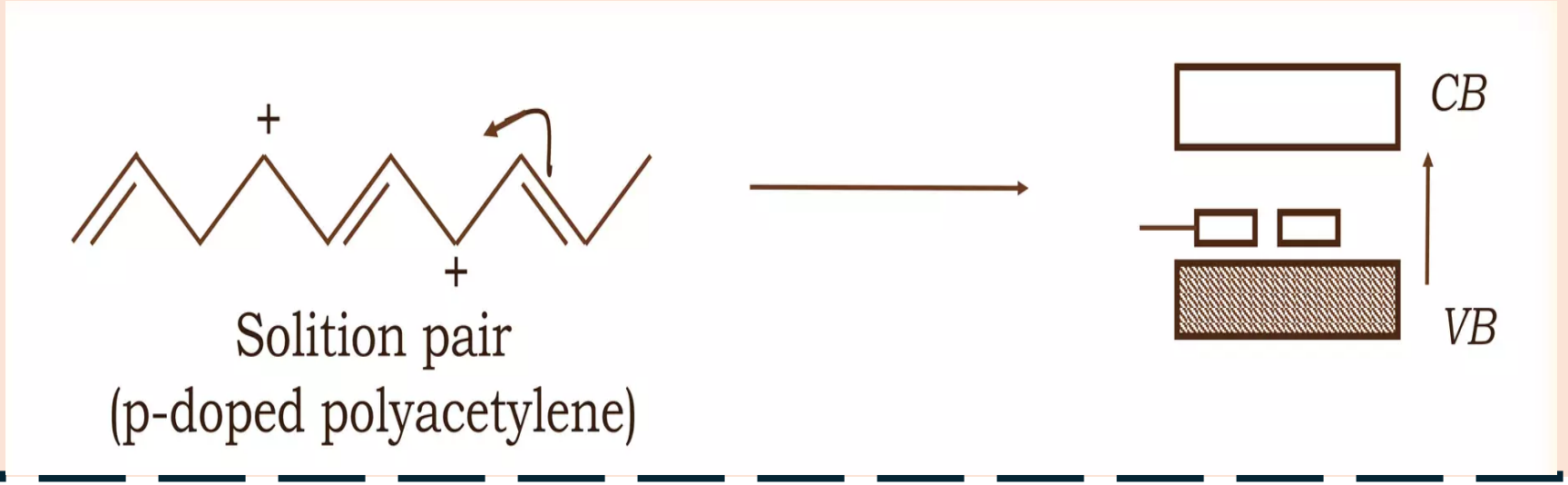
A bipolaron is a quasiparticle that forms when two polarons, typically two electrons or two holes, become bound together in a solid material.



p-doping

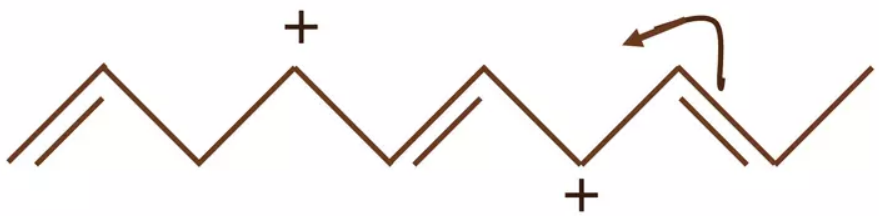


The bipolaron is represented by the paired energy levels, where both levels are occupied within the band gap.

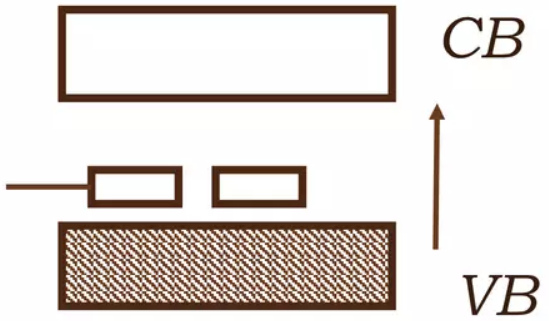


In p-doped polyacetylene, doping introduces holes (positive charges) into the polymer chain, which can lead to the formation of solitons.

p-doping



Solition pair
(p-doped polyacetylene)

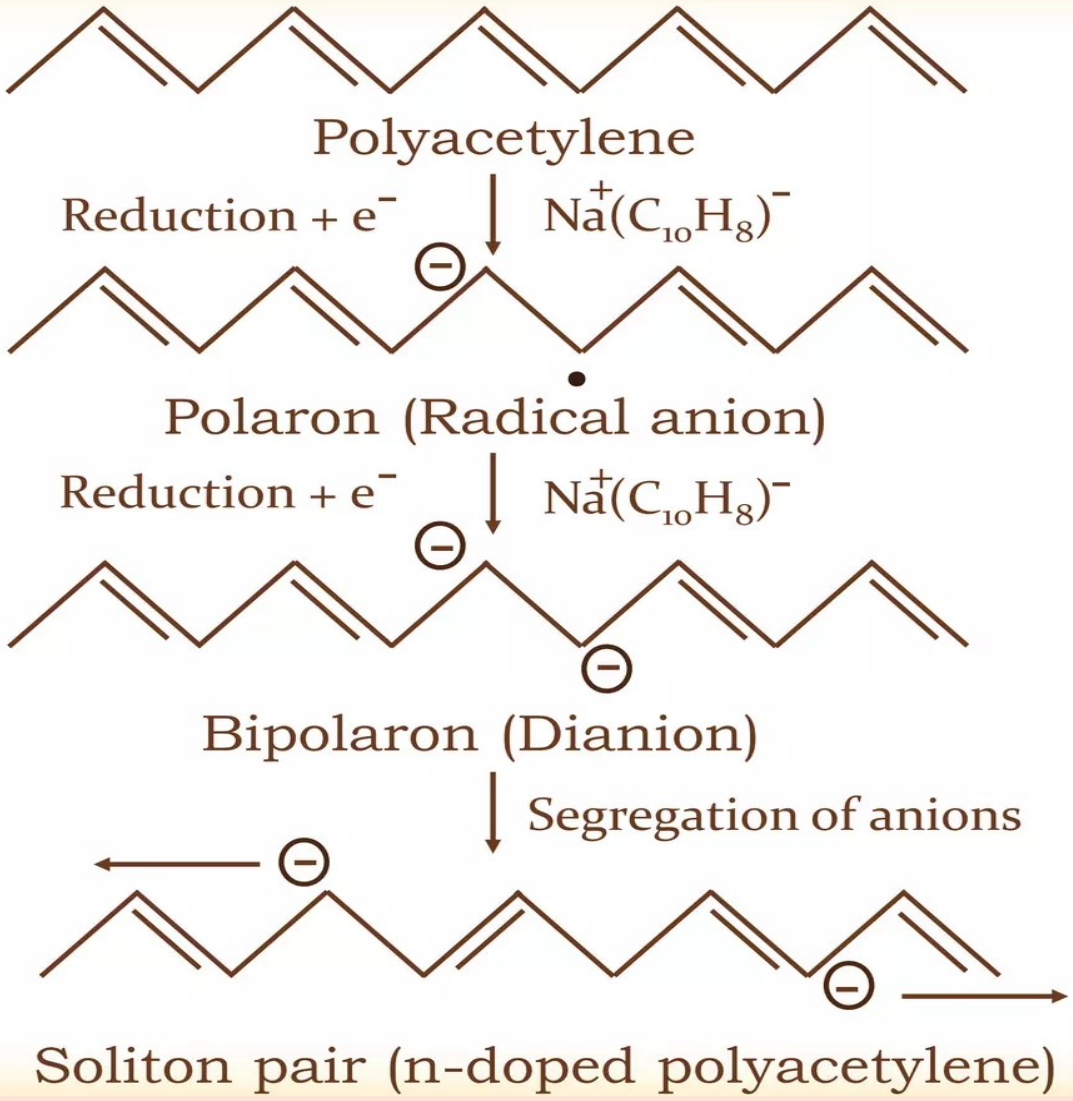


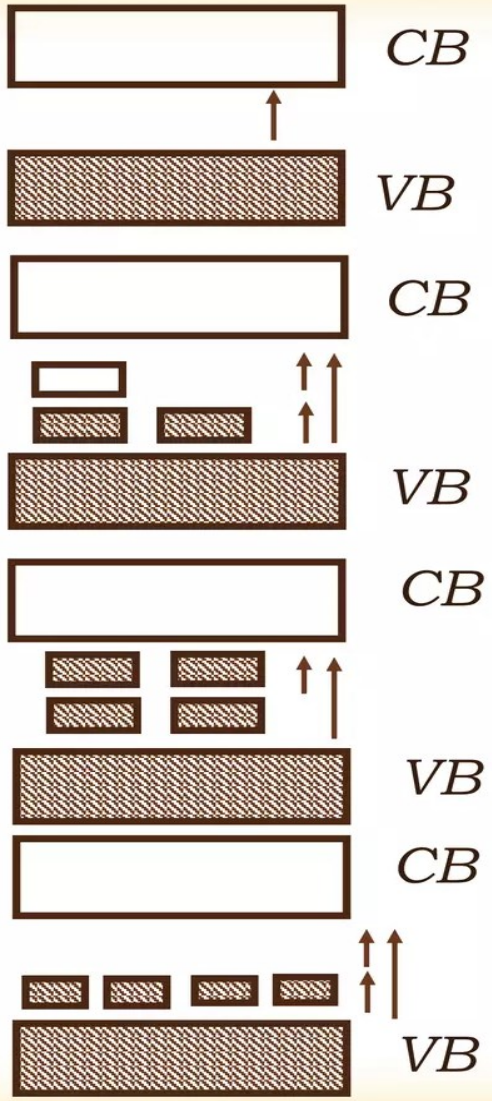
n-doping

- This is done by reduction process
- For this conductance, polymer having conjugation is treated with Lewis's base like sodium naphthalide

$$-(CH{=}CH)_n - + Na^{\oplus}(C_{10}H_8)^{\ominus} \longrightarrow Na^{\oplus}-(CH{=}CH)_n^{\ominus} + C_{10}H_8$$

n-doping


Polyacetylene
Reduction + e⁻
Na⁺(C₁₀H₈)⁻
Polaron (Radical anion)
Reduction + e⁻
Na⁺(C₁₀H₈)⁻
Bipolaron (Dianion)
Segregation of anions
Soliton pair (n-doped polyacetylene)


CB
VB
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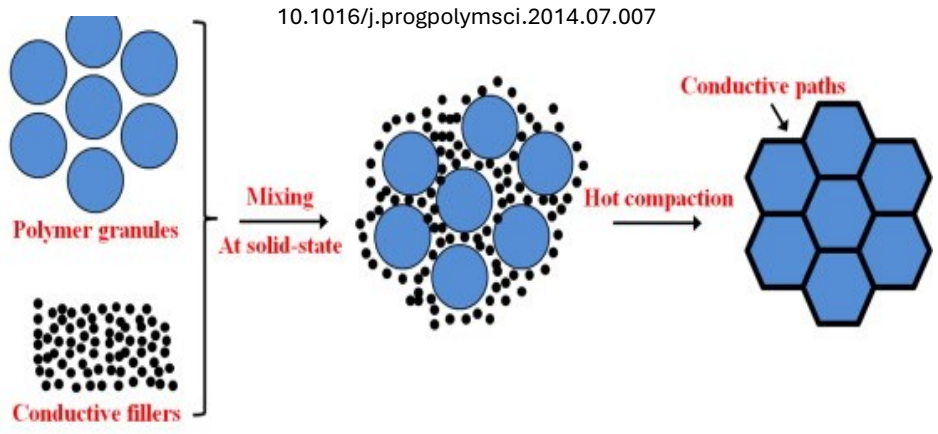
Extrinsically CONDUCTING POLYMERS (ECPs)

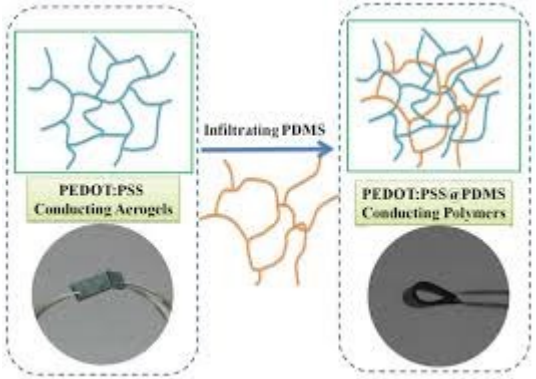
- These polymers possess their conductivity due to the presence of externally added ingredients in them.
- These are of two types:

Classification

Conductive element filled polymers

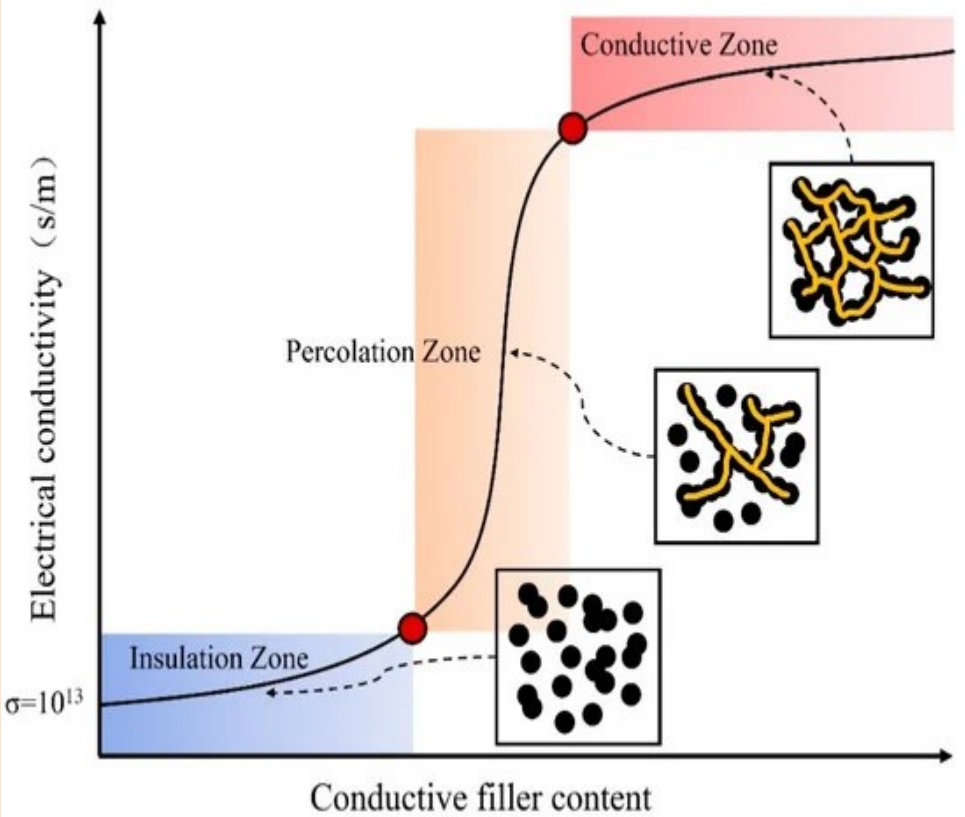


Blended conducting polymers



1. Conductive element filled polymers

- The polymer acts as the binder to hold the conducting elements (such as carbon black, metallic fibers, metallic oxides etc.) together in the solid entity.
- Minimum concentration of conductive filler, which should be added so that polymer starts conducting is known as **percolation threshold**.
- Because at this concentration of filler or conducting element, a conducting path is formed in polymeric material.
- Such compounds have been important, for example in hospital operation theatres where it was essential that static charges did not build up, leading to explosion involving anesthetics.



2. Blended conducting polymers

- These polymers can be obtained by blending processes.
- They possesses better physical, chemical, electrical and mechanical properties and can be easily processed.
- E.g: upto 40% of polypyrrole will have little effect on tensile strength and give much higher impact strength. Such compounds are of interest in electromagnetic shielding.

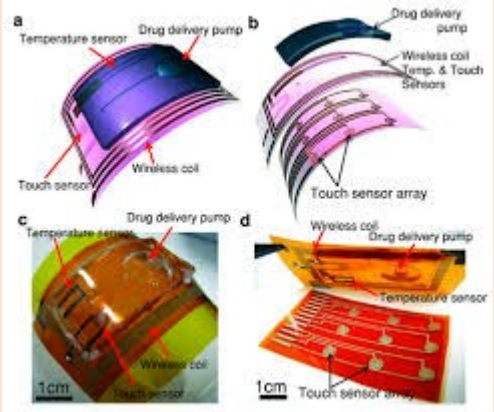
Applications of CONDUCTING POLYMERS

❑ Engineering applications of conducting polymers:

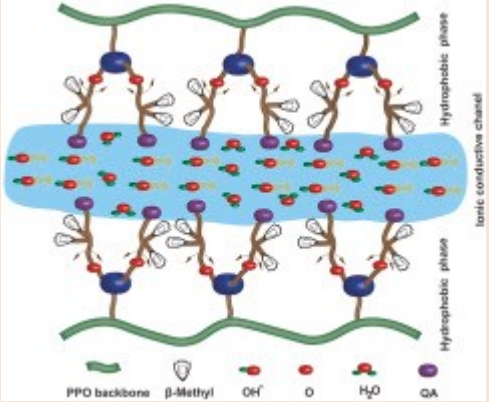
There are several utilities of conducting polymer due to their better physical, chemical, mechanical properties, lightweight and easy to process.

Some of them are:

To make rechargeable, lightweight batteries

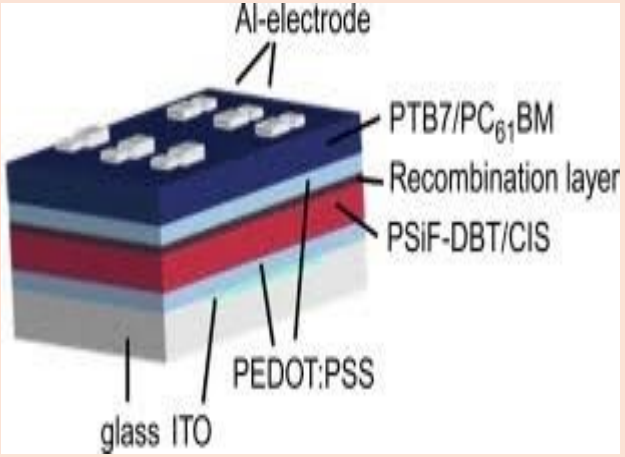


In making of analytical sensors for pH O_2, NO_2, SO_2, glucose

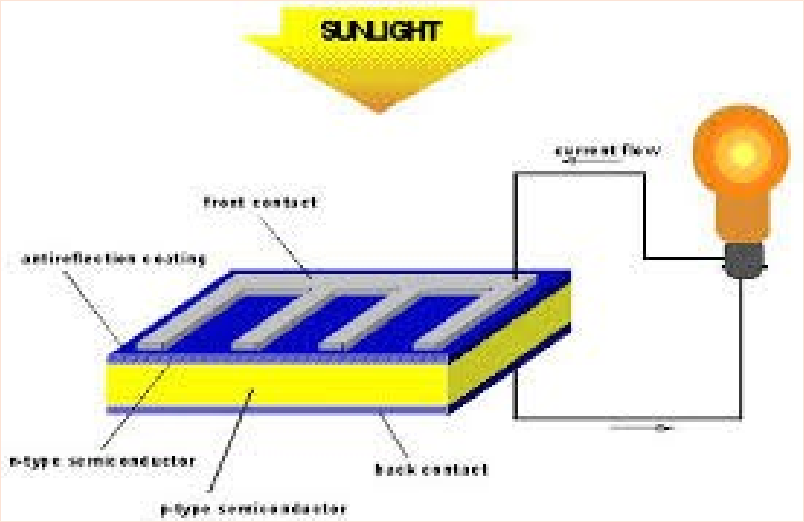


making of Ion exchangers

Applications of CONDUCTING POLYMERS



Making of solar cells



In photovoltaic devices